



OR 6205: Deterministic Operations Research
Spring 2026

Deterministic Operations Research

Week 3 – Class 5: Solving Linear Programming models using SIMPLEX algorithm

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Objectives

Classify a solution of a LP as basic versus non-basic and feasible versus infeasible.

Derive the basic solution associated with a corner point solution

Describe the three steps of a simplex iteration and demonstrate them graphically with corner points

Apply the simplex algorithm in a tabular form to solve a small LP, manually and interactively in the computer using the IOR tutorial

Identify, in an optimal simplex tableau, the values of all variables, the shadow prices and the reduced costs and describe their meaning

Identify the binding (active) constraints of the solution of a LP

Identify any redundant constraints of a LP with two variables graphically

Due exercises

Homework 1 – Sunday

Upload in-class exercise – before class on Tuesday (01/27)

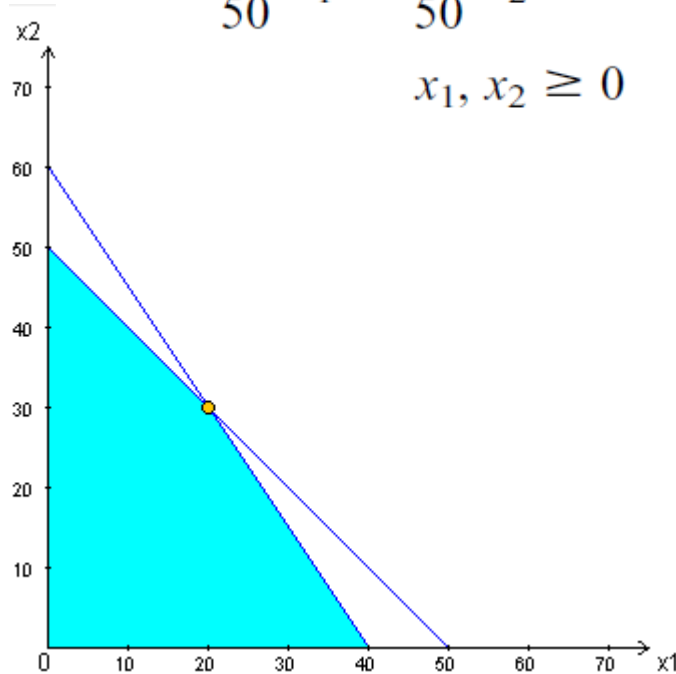
Graphical Technique: In-class exercise

$$\max z = 3x_1 + 2x_2$$

$$\text{s.t.} \quad \frac{1}{40}x_1 + \frac{1}{60}x_2 \leq 1$$

$$\frac{1}{50}x_1 + \frac{1}{50}x_2 \leq 1$$

$$x_1, x_2 \geq 0$$



$$\max z = 3x_1 + 2x_2$$

$$\text{s.t.} \quad \frac{1}{40}x_1 + \frac{1}{60}x_2 \leq 1$$

$$\frac{1}{50}x_1 + \frac{1}{50}x_2 \leq 1$$

$$x_1 \geq 30$$

$$x_2 \geq 20$$

$$x_1, x_2 \geq 0$$

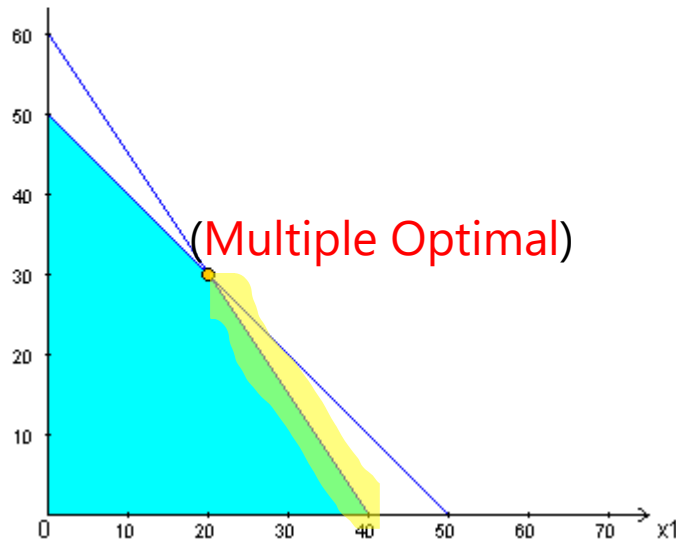
$$\min z = 3x_1 + 2x_2$$

$$\text{s.t.} \quad \frac{1}{40}x_1 + \frac{1}{60}x_2 \leq 1$$

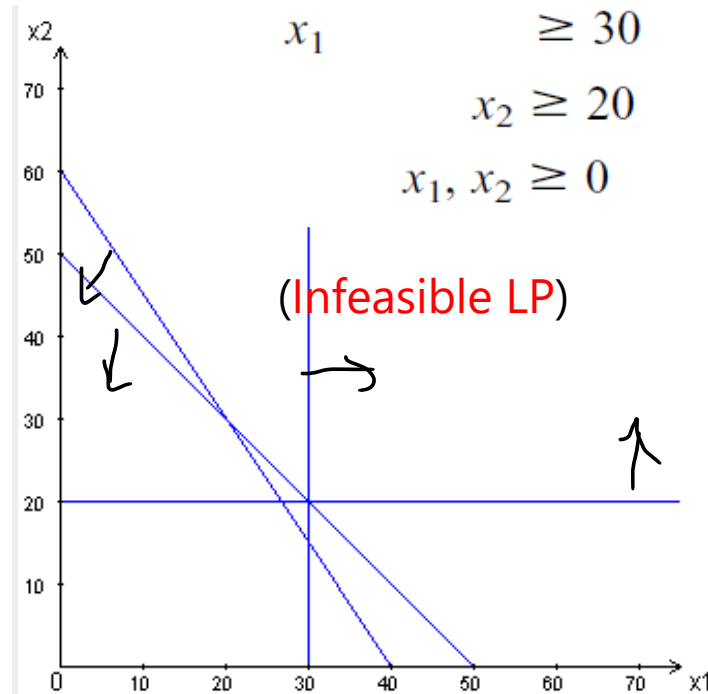
$$\frac{1}{50}x_1 + \frac{1}{50}x_2 \leq 1$$

Graphical Technique: In-class exercise

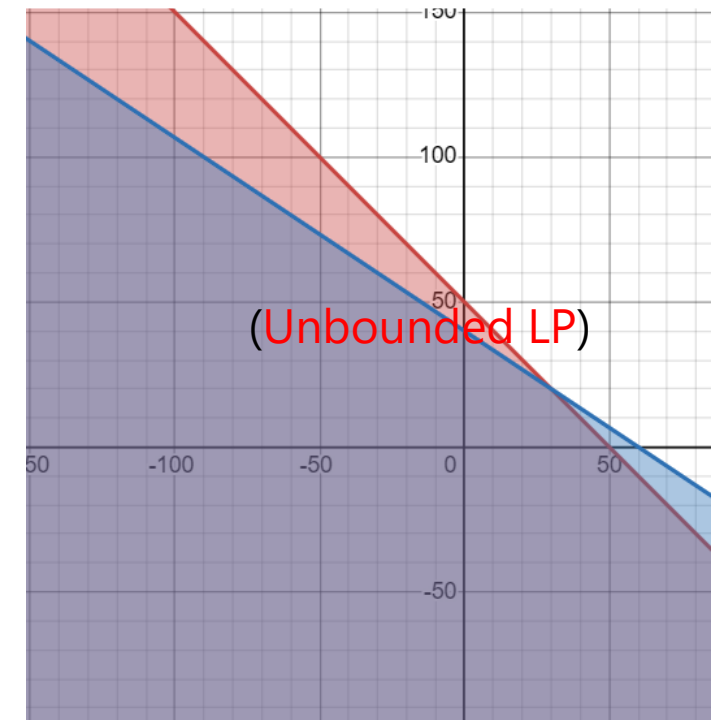
$$\begin{aligned} \max z &= 3x_1 + 2x_2 \\ \text{s.t.} \quad &\frac{1}{40}x_1 + \frac{1}{60}x_2 \leq 1 \\ &\frac{1}{50}x_1 + \frac{1}{50}x_2 \leq 1 \\ &x_1, x_2 \geq 0 \end{aligned}$$



$$\begin{aligned} \max z &= 3x_1 + 2x_2 \\ \text{s.t.} \quad &\frac{1}{40}x_1 + \frac{1}{60}x_2 \leq 1 \\ &\frac{1}{50}x_1 + \frac{1}{50}x_2 \leq 1 \end{aligned}$$



$$\begin{aligned} \min z &= 3x_1 + 2x_2 \\ \text{s.t.} \quad &\frac{1}{40}x_1 + \frac{1}{60}x_2 \leq 1 \\ &\frac{1}{50}x_1 + \frac{1}{50}x_2 \leq 1 \end{aligned}$$



Introduction to Simplex

Overview

George Dantzig's development of the simplex method made it possible to systematically solve incredibly complex problems.

Simplex method, combined with powerful computer applications that perform the calculations, have made the use of LP applicable in many different fields.

Correct use of the simplex method gives you the ability to construct and answer complex questions consisting of **100s/1000s of dimensions** and makes you an invaluable member of a project team.

From its wartime origins to modern-day applications, the **simplex method** remains a **fundamental optimization tool**.

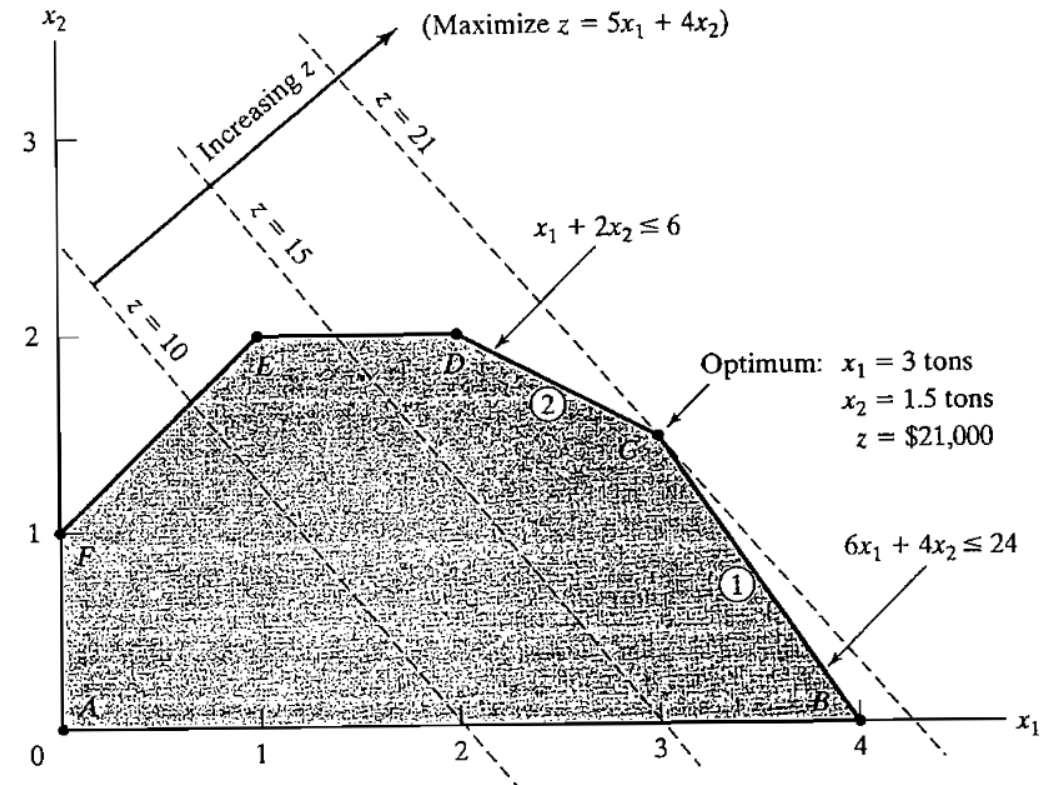
Overview

The simplex method is a **powerful algorithm** used to solve linear programming problems

Can be used to solve small LP problems **by hand**.

The method **systematically moves along the edges of the feasible region**, which is defined by the constraints, to find the optimal solution.

This **iterative process** continues until no further improvements are possible, resulting in the optimal solution.



LP as a system of Linear Equations

Gaussian elimination, also known as **row reduction**, is an **algorithm** for solving **systems of linear equations**

Using these operations, a matrix can always be transformed into *reduced row echelon form*: each nonzero row is above every zero row, each nonzero row has leftmost nonzero entry equal to 1, the columns containing these leading 1s have all other entries 0, and the leading 1 in each nonzero row is to the right of the leading 1 in the previous row. This final form is unique; in other words, it is independent of the sequence of row operations used.

Using row operations to convert a matrix into reduced row echelon form is sometimes called **Gauss–Jordan elimination**.

$$\begin{bmatrix} 1 & 3 & -1 \\ 0 & 1 & 7 \end{bmatrix} \xrightarrow{\text{subtract } 3 \times (\text{row } 2) \text{ from row } 1} \begin{bmatrix} 1 & 0 & -22 \\ 0 & 1 & 7 \end{bmatrix}$$

Simplex Algorithm uses GJ elimination

LP as a system of Linear Equations – Standard Form

LP is a system of inequalities. For us to perform the algorithm, the initial LP model needs to follow a set of rules/format.

- a. adding slacks to the original inequality constraints to convert them to equalities (augmented form)
- b. In addition to the original variables, an augmented solution includes the slacks

Standard Form

An LP is in standard form if the following are true:

- All variables have **non-negativity** constraints.
- All other constraints are strict **equality constraints**.
- The **right-hand side** of all constraints are non-negative.
- The objective function can be max/min.

The following LP is not in standard form:

maximize: $z = 3x_1 + 2x_2 - x_3 + x_4$

subject to: $x_1 + 2x_2 + x_3 - x_4 \leq 5$

$$-2x_1 - 4x_2 + x_3 + x_4 \leq -1$$

$$x_1 \geq 0$$

$$x_2 \geq 0$$

How do we convert it to standard form?

Converting Inequalities to Equalities

How do you convert $\leq$ to an equality?

- Add a **non-negative slack variable** to the LHS.

For example, the following inequality:

$$x_1 + 2x_2 + x_3 - x_4 \leq 5$$

becomes the following equality:

$$x_1 + 2x_2 + x_3 - x_4 + s_1 = 5$$

- note we also have $s_1 \geq 0$

The slack variable measures the amount of unused resource.

- We can also say that:

$$s_1 = -x_1 - 2x_2 - x_3 + x_4 + 5$$

How do you convert $\geq$ to an equality?

- Subtract a **non-negative surplus variable** to the LHS.
- For example, if we had the following inequality:

$$x_1 + 2x_2 + x_3 - x_4 \geq 5$$

- becomes the following equality:

$$x_1 + 2x_2 + x_3 - x_4 - e_1 = 5$$

- note we also have $e_1 \geq 0$

The surplus variable measures the amount by which we over-satisfy a requirement.

- So we can also say that:

$$e_1 = x_1 + 2x_2 + x_3 - x_4 - 5$$

Converting Minimization to Maximization

How do you convert a minimization problem to a maximization problem?

Converting Minimization to Maximization

How do you convert a minimization problem to a maximization problem?

Recognize that to minimize z , you could alternatively **maximize the negative of z**

- For example, the following minimization objective function:

$$\min \quad x_1 + 2x_2 + x_3 - x_4$$

- becomes the following maximization objective function:

$$\max \quad -x_1 - 2x_2 - x_3 + x_4$$

Dealing with Variables Unrestricted in Sign

How do you deal with a variable unrestricted in sign?

- For example, the following LP:

$$\begin{aligned} \max \quad & -x_1 - 2x_2 - x_3 + x_4 \\ & x_1 + 3x_2 + 4x_3 - x_4 \leq 5 \\ & x_1 \geq 0, \textbf{x}_2 \text{ unrestricted}, x_3 \geq 0, x_4 \geq 0 \end{aligned}$$

Replace it by a pair of new non-negative variables:

x_a and x_b , where $x = x_a - x_b$.

- becomes the following new LP:

$$\begin{aligned} \max \quad & -x_1 - 2(x_{2a} - x_{2b}) - x_3 + x_4 \\ & x_1 + 3(x_{2a} - x_{2b}) + 4x_3 - x_4 \leq 5 \\ & x_{2a} \geq 0, x_{2b} \geq 0 \\ & x_1 \geq 0, x_3 \geq 0, x_4 \geq 0 \end{aligned}$$

We convert the original variable by inspection after solution of the new LP.

Dealing with Non-Positive Variables

How do you deal with a non-positive (e.g., $x \leq 0$) variable?

- Replace it by a new non-negative variable that is its negative.
- For example, the following LP:

$$\max \quad -x_1 - 2x_2 - x_3 + x_4$$

$$x_1 + 3x_2 + 4x_3 - x_4 \leq 5$$

$$x_1 \geq 0, x_2 \leq 0, x_3 \leq 0, \mathbf{x_4 \geq 0}$$

- becomes the following new LP:

$$\max \quad -x_1 + 2y_2 + y_3 + x_4$$

$$x_1 - 3y_2 - 4y_3 - x_4 \leq 5$$

$$x_1 \geq 0, y_2 \geq 0, y_3 \geq 0, x_4 \geq 0$$

We convert the original variable by inspection after solution of the new LP.

In-Class Problem – Standard Form Conversion

$$\text{Minimize } Z = x_1 - x_2 + 2x_3 - x_4$$

$$\begin{array}{lcl} \text{Subject to:} & 2x_1 - 3x_2 + x_3 & \geq 8 \\ & 2x_1 + 2x_2 + 3x_3 - x_4 & \leq 2 \\ & 4x_1 + 4x_2 + 2x_3 - x_4 & \geq 6 \\ & x_1 - 2x_2 + 3x_3 - x_4 & \leq 2 \\ & x_1 & \geq 0 \\ & x_2 & \leq 0 \\ & x_4 & \geq 0 \end{array}$$

In-Class Problem – Standard Form Conversion

$$\text{Minimize } Z = x_1 - x_2 + 2x_3 - x_4$$



$$\text{Maximize } -Z = -x_1 - y_2 - 2x_3^+ + 2x_3^- + x_4$$

$$\begin{aligned} \text{Subject to: } & 2x_1 - 3x_2 + x_3 \geq 8 \\ & 2x_1 + 2x_2 + 3x_3 - x_4 \leq 2 \\ & 4x_1 + 4x_2 + 2x_3 - x_4 \geq 6 \\ & x_1 - 2x_2 + 3x_3 - x_4 \leq 2 \\ & x_1 \geq 0 \\ & x_2 \leq 0 \\ & x_4 \geq 0 \end{aligned}$$

$$\begin{aligned} \text{Subject to: } & 2x_1 + 3y_2 + x_3^+ - x_3^- - e_1 = 8 \\ & 2x_1 - 2y_2 + 3x_3^+ - 3x_3^- - x_4 + s_1 = 2 \\ & 4x_1 - 4y_2 + 2x_3^+ - 2x_3^- - x_4 - e_2 = 6 \\ & x_1 + 2y_2 + 3x_3^+ - 3x_3^- - x_4 + s_2 = 2 \\ & x_1, x_3^+, x_3^-, x_4, y_2, e_1, e_2, s_1, s_2 \geq 0 \end{aligned}$$

Introduction to Matrix Operations

for LP solver

Linear Systems

Consider this system of linear equations:

$$\begin{array}{ccccccc} a_{11}x_1 & + & a_{12}x_2 & + & \cdots & + & a_{1n}x_n & = & b_1 \\ a_{21}x_1 & + & a_{22}x_2 & + & \cdots & + & a_{2n}x_n & = & b_2 \\ \vdots & & \vdots & & & & \vdots & & \\ a_{m1}x_1 & + & a_{m2}x_2 & + & \cdots & + & a_{mn}x_n & = & b_m \end{array}$$

Here, $x_1, x_2, \dots, x_n$ are **variables**, and the a_{ij} 's and b_i 's are **constants**.

Linear Systems

Consider this system of linear equations:

$$\begin{array}{ccccccc} a_{11}x_1 & + & a_{12}x_2 & + & \cdots & + & a_{1n}x_n & = & b_1 \\ a_{21}x_1 & + & a_{22}x_2 & + & \cdots & + & a_{2n}x_n & = & b_2 \\ \vdots & & \vdots & & & & \vdots & & \\ a_{m1}x_1 & + & a_{m2}x_2 & + & \cdots & + & a_{mn}x_n & = & b_m \end{array}$$

This set of equations is called a linear system of m equations in n variables.

Here, $x_1, x_2, \dots, x_n$ are variables, and the a_{ij} 's and b_i 's are constants.

Using Matrices to represent LPs

- ! Using matrices can **greatly simplify** the statement and solution of a system of linear equations.

● The linear system

$A\mathbf{x} = \mathbf{b}$, where:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ \vdots & \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

can be represented as

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

In-Class Exercise

Consider the linear program

$$\begin{array}{ll}\min & 9x_1 + 12x_2 + 6x_3 \\ \text{s.t.} & 3x_1 + x_2 - 2x_3 \geq 4 \\ & x_1 + 2x_2 + 4x_3 \geq 6 \\ & x_j \geq 0.\end{array}$$

$$\begin{array}{ll}\min & \mathbf{c}^T \mathbf{x} \\ \text{s.t.} & \mathbf{Ax} \geq \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0}.\end{array}$$

How to convert this to matrix form?

In-Class Exercise

Converting the objective function and constraints to matrix form, we have:

$$\begin{aligned} \min \quad & [9 \quad 12 \quad 6] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \\ \text{s.t.} \quad & \begin{bmatrix} 3 & 1 & -2 \\ 1 & 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \geq \begin{bmatrix} 4 \\ 6 \end{bmatrix} \\ & x_j \geq 0. \end{aligned}$$

If we let

$$\mathbf{A} = \begin{bmatrix} 3 & 1 & -2 \\ 1 & 2 & 4 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 4 \\ 6 \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} 9 \\ 12 \\ 6 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

then this linear program can be written as:

$$\begin{aligned} \min \quad & \mathbf{c}^T \mathbf{x} \\ \text{s.t.} \quad & \mathbf{Ax} \geq \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0}. \end{aligned}$$

Steps of the Simplex Algorithm

Step 1: Formulate the Linear Programming Problem

Step 2: Convert to Standard Linear Programming Form

Step 3: Initialize the Simplex Tableau

Step 4: Start at a corner if the feasible region

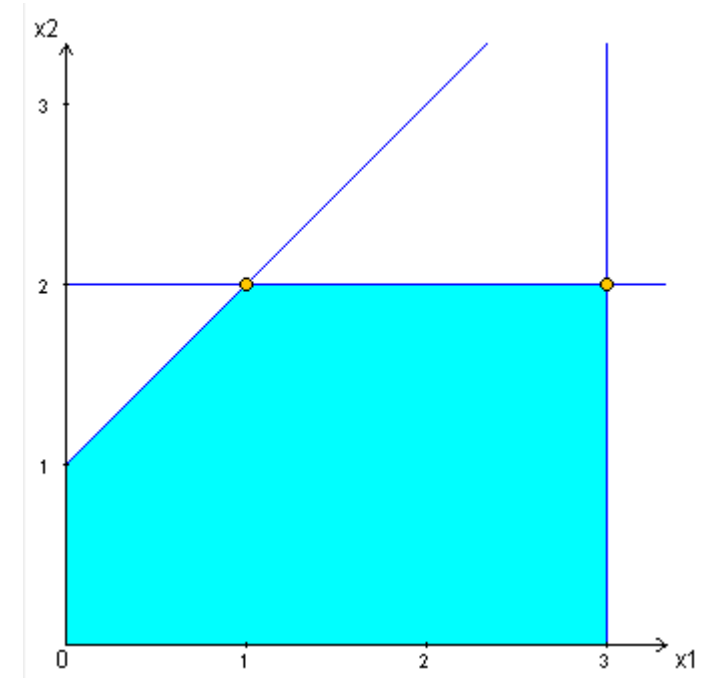
Step 5: Follow the path and look for an edge that takes you closer to your goal

Step 6: Stop when you can't find an edge that improves your goal! ***You're now at the best corner -the optimal solution!***

Steps of Simplex algorithm

Step 1: Formulate the LP

We will work on this small LP : Maximize $x_1 + x_2$
subject to $-x_1 + x_2 \leq 1$
 $x_1 \leq 3$
 $x_2 \leq 2$
 $x_1, x_2 \geq 0$.



Steps of Simplex algorithm

Step 1: Formulate the LP

We will work on this small LP :

$$\begin{array}{ll} \text{Maximize} & x_1 + x_2 \\ \text{subject to} & -x_1 + x_2 \leq 1 \\ & x_1 \leq 3 \\ & x_2 \leq 2 \\ & x_1, x_2 \geq 0. \end{array}$$

Step 2: Introduce slack variable x_3, x_4, x_5 and represent the model in standard LP form

$$\begin{array}{ll} \text{maximize} & x_1 + x_2 \\ \text{subject to} & -x_1 + x_2 + x_3 = 1 \\ & x_1 + x_4 = 3 \\ & x_2 + x_5 = 2 \\ & x_1, x_2, \dots, x_5 \geq 0, \end{array}$$

Steps of Simplex algorithm

Step 1: Formulate the LP

We will work on this small LP : **Maximize** $x_1 + x_2$
subject to $-x_1 + x_2 \leq 1$
 $x_1 \leq 3$
 $x_2 \leq 2$
 $x_1, x_2 \geq 0.$

$$A = \begin{pmatrix} & x_1 & x_2 \\ -1 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Step 2: Introduce slack variable x_3, x_4, x_5 and represent the model in standard LP form

$$\begin{array}{ll} \text{maximize} & x_1 + x_2 \\ \text{subject to} & -x_1 + x_2 + x_3 = 1 \\ & x_1 + x_4 = 3 \\ & x_2 + x_5 = 2 \\ & x_1, x_2, \dots, x_5 \geq 0, \end{array}$$

$$A = \begin{pmatrix} & x_1 & x_2 & x_3 & x_4 & x_5 \\ -1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{pmatrix}$$



Steps of Simplex algorithm

Step 3: Express each linear program in the form of a *simplex tableau*

$$\begin{array}{rclcl} x_3 & = & 1 & + & x_1 & - & x_2 \\ x_4 & = & 3 & - & x_1 & & \\ x_5 & = & 2 & & & - & x_2 \\ \hline z & = & & & x_1 & + & x_2 \end{array}$$

Steps of Simplex algorithm

Step 3: Express each linear program in the form of a *simplex tableau*

Slack Variables

$$\begin{array}{rcl}
 x_3 & = & 1 + x_1 - x_2 \\
 x_4 & = & 3 - x_1 \\
 x_5 & = & 2 \qquad - x_2 \\
 \hline
 z & = & \qquad x_1 + x_2
 \end{array}$$

IOR Tutorial – Simplex in Tableau form

Bas ...	Eq.	Z	x1	x2	x3	x4	x5	Right Side
Z	0	1	-1	-1	0	0	0	0
x3	1	0	-1	1	1	0	0	1
x4	2	0	1	0	0	1	0	3
x5	3	0	0	1	0	0	1	2

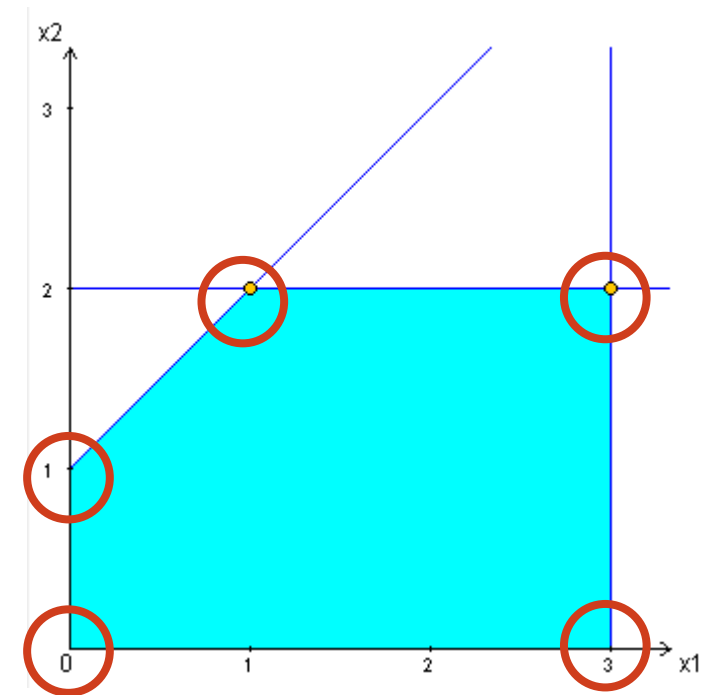
Steps of Simplex algorithm

Step 4: Select a corner of the feasible region

$$\begin{array}{rcl}
 x_3 & = & 1 + x_1 - x_2 \\
 x_4 & = & 3 - x_1 \\
 x_5 & = & 2 - x_2 \\
 \hline
 z & = & x_1 + x_2
 \end{array}$$

What would be a good starting point? Why?

Corner feasible points



Steps of Simplex algorithm

Step 4: Select a corner of the feasible region

$$\begin{array}{rcl}
 x_3 & = & 1 + x_1 - x_2 \\
 x_4 & = & 3 - x_1 \\
 x_5 & = & 2 \qquad \qquad - x_2 \\
 \hline
 z & = & \qquad x_1 + x_2
 \end{array}$$

Identity matrix 

$$A = \begin{pmatrix} x_1 & x_2 & x_3 & x_4 & x_5 \\ -1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{pmatrix}$$

We substitute x_1 , and x_2 with 0 then,
 $x_3 = 1$, $x_4 = 3$, $x_5 = 2$ and $z = 0$

Steps of Simplex algorithm

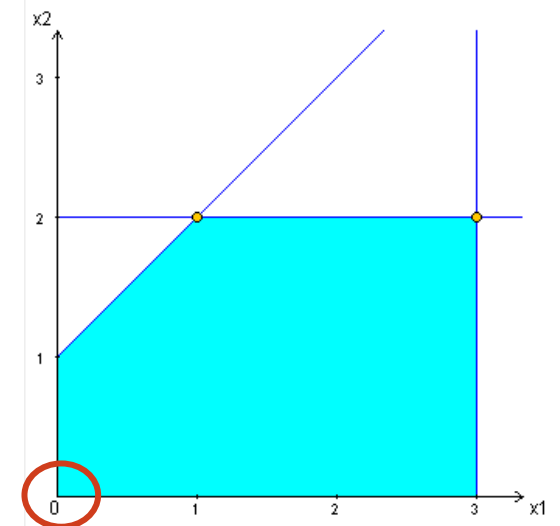
Step 4 cont'd: Select a corner of the feasible region

So, $x_1 = 0$, $x_2 = 0$, $x_3 = 1$, $x_4 = 3$, $x_5 = 2$ is a Basic Feasible Solution (BFS) where,

Non-basic variables (NBV) $\xrightarrow{\quad}$ $\xleftarrow{\quad}$ Basic variables (BV)

$$A = \begin{pmatrix} x_1 & x_2 & x_3 & x_4 & x_5 \\ -1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{pmatrix}$$

Our Basis is $\mathbf{B} = \{3, 4, 5\}$



What would be the equivalent result in the original (non augmented model)?

Steps of Simplex algorithm

Step 4 cont'd: Select a corner of the feasible region

So, $x_1 = 0$, $x_2 = 0$, $x_3 = 1$, $x_4 = 3$, $x_5 = 2$ is a Basic Feasible Solution (BFS) where,

Non-basic variables (NBV) $\curvearrowright$

$$A = \begin{pmatrix} x_1 & x_2 & x_3 & x_4 & x_5 \\ -1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{pmatrix}$$

Basic variables (BV) $\curvearrowright$

Our Basis is $\mathbf{B} = \{3, 4, 5\}$

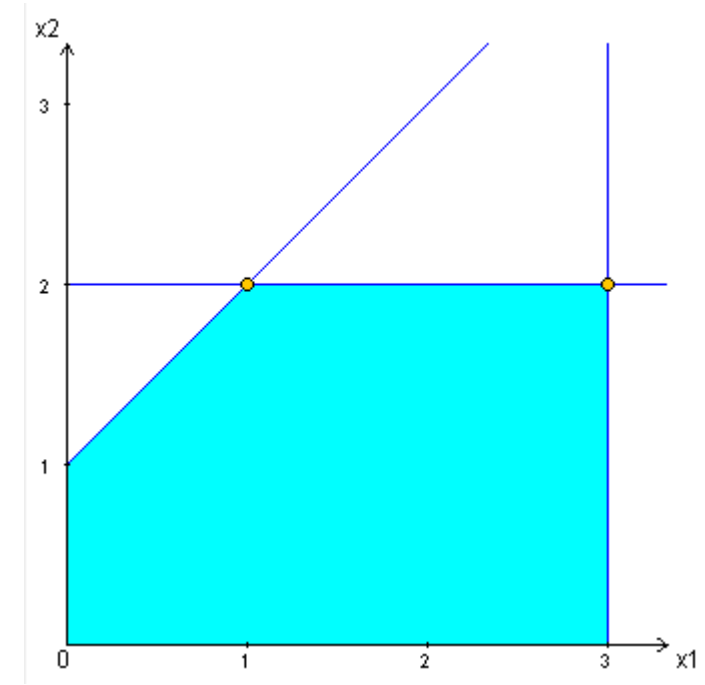
Is this answer the optimal answer?

Steps of Simplex algorithm

Step 5: Find the entering variable that will improve Z

What would be a good option? What criteria?

$$\begin{array}{llllll} \text{maximize} & x_1 & + & x_2 & & \\ \text{subject to} & -x_1 & + & x_2 & + & x_3 & = & 1 \\ & x_1 & & & + & x_4 & = & 3 \\ & & & x_2 & & & + & x_5 & = & 2 \\ & x_1, x_2, \dots, x_5 & \geq & 0, \end{array}$$



Steps of Simplex algorithm

Step 5: Find the entering variable that will improve Z

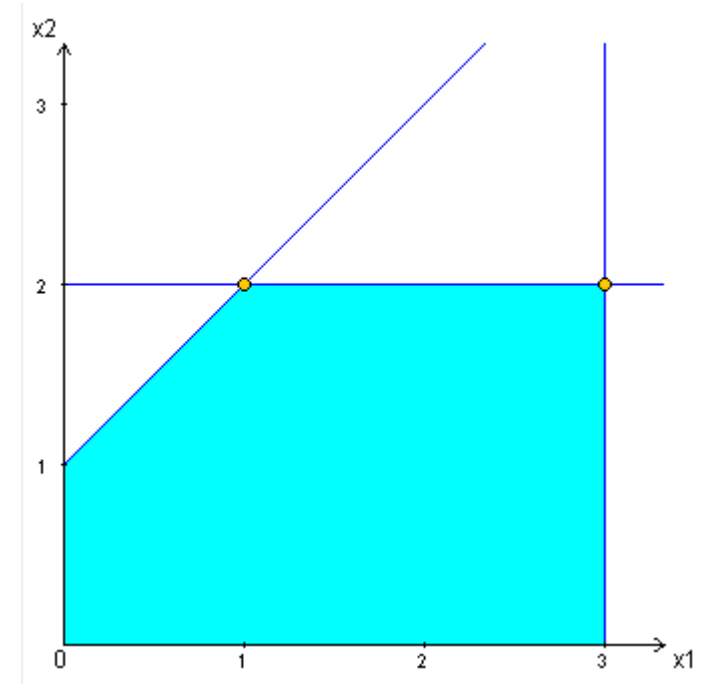
Pick a variable arbitrarily from NBVs (x_1, x_2) to improve from 0 because both variables have positive coefficients in the objective

Let's pick x_2

$$\begin{array}{llllll} \text{maximize} & x_1 & + & x_2 & & \\ \text{subject to} & -x_1 & + & x_2 & + & x_3 & = & 1 \\ & x_1 & & & + & x_4 & = & 3 \\ & & & x_2 & & & + & x_5 & = & 2 \\ & & & & & & & & & x_1, x_2, \dots, x_5 \geq 0, \end{array}$$



How much can you increase x_2 ?



Steps of Simplex algorithm

Step 5: Find the entering variable that will improve Z

Pick a variable arbitrarily from NBVs (x_1, x_2) to improve from 0 because both variables have positive coefficients in the objective

Let's pick x_2

$$\begin{array}{llll} \text{maximize} & x_1 + x_2 & & \\ \text{subject to} & -x_1 + x_2 + x_3 & = & 1 \\ & x_1 & + & x_4 = 3 \\ & & x_2 & + x_5 = 2 \\ & x_1, x_2, \dots, x_5 & \geq & 0, \end{array}$$



How much can you increase x_2 ?

It depends on what?

Equivalent step to optimality test and entering variable from textbook

Steps of Simplex algorithm

Step 5 cont'd: Find the leaving variable that will improve Z

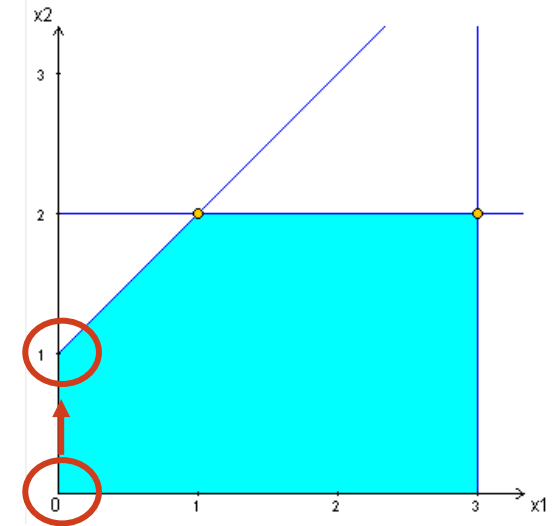
! While increasing x_2 , be careful not to let any of the basic variables x_3, x_4, x_5 go below zero!

Let's check our BVs:

$$x_3 = 1 + x_1 - x_2$$

$$x_4 = 3 - x_1$$

$$x_5 = 2 - x_2$$



Steps of Simplex algorithm

Step 5 cont'd: Find the leaving variable that will improve Z

! While increasing x_2 , be careful not to let any of the basic variables x_3, x_4, x_5 go below zero!

Let's check our BVs:

$$\begin{array}{lcl}
 x_3 = 1 + x_1 - x_2 & \xrightarrow{\quad} & x_2 = 1 \\
 x_4 = 3 - x_1 & & \\
 x_5 = 2 - x_2 & \xrightarrow{\quad} & x_2 = 2
 \end{array}$$

We can only increase x_2 up to 1 without making other variables go below zero!

Equivalent steps to min ratio test from the textbook

Steps of Simplex algorithm

and leaving

Step 5 cont'd: Find the entering variable that will improve Z

IOR Tutorial – Simplex in Tableau form

Bas ...	Eq.	Z	x1	x2	x3	x4	x5	Right Side
Z	0	1	-1	-1	0	0	0	0
x3	1	0	-1	1	1	0	0	1
x4	2	0	1	0	0	1	0	3
x5	3	0	0	1	0	0	1	2

Entering variable = x_2
Leaving variable = x_3

Make Elementary Row Operations (ERO) to transform all values highlighted cells to 0, but the pivot (the highlighted intersected cell) that would be 1

Steps of Simplex algorithm

and leaving

Step 5 cont'd: Find the entering variable that will improve z

The entering variable $x_2 = 1$ while, $x_1 = 0$, $x_3 = 0$, $x_4 = 3$, $x_5 = 1$

$$x_3 = 1 + x_1 - x_2$$

$$x_4 = 3 - x_1$$

$$x_5 = 2 - x_2$$

What is the new basis? And z ?

Steps of Simplex algorithm

Step 5 cont'd: Find the entering variable that will improve z

The entering variable $x_2 = 1$ while, $x_1 = 0$, $x_3 = 0$, $x_4 = 3$, $x_5 = 1$

$$x_3 = 1 + x_1 - x_2$$

$$x_4 = 3 - x_1$$

$$x_5 = 2 - x_2$$

Basis becomes $B = \{2, 4, 5\}$, **and** $z = 1$

Steps of Simplex algorithm

Step 6: Update the simplex tableau

$$\begin{array}{rcl} x_2 & = & 1 + x_1 - x_3 \\ x_4 & = & 3 - x_1 \\ x_5 & = & 1 - x_1 + x_3 \\ \hline z & = & 1 + 2x_1 - x_3 \end{array}$$

The process of rewriting one simplex tableau into another is called a **pivot step**!

Equivalent steps to the Gauss-Jordan elementary row operations that you've seen in the textbook!

Steps of Simplex algorithm

Repeat Step 5: Find the entering variable that will improve z

The objective becomes $z = 1 + 2x_1 - x_3$

Only improving x_1 will improve z ; thus, the entering variable is x_1



How much can you increase x_1 ?

Steps of Simplex algorithm

Repeat Step 5: Find the entering variable that will improve z

The objective becomes $z = 1 + 2x_1 - x_3$

Only improving x_1 will improve z ; thus, the entering variable is x_1

? How much can you increase x_1 ?

$$\begin{array}{rcl}
 x_2 & = & 1 + x_1 - x_3 \\
 x_4 & = & 3 - x_1 \\
 x_5 & = & 1 - x_1 + x_3 \\
 \hline
 z & = & 1 + 2x_1 - x_3
 \end{array}$$

Does not restrict!

$x_1 = 3$

$x_1 = 1$

Steps of Simplex algorithm

Step 5 cont'd: Find the entering variable that will improve z

The entering variable $x_1 = 1$ while, $x_2 = 2$, $x_3 = 0$, $x_4 = 2$, $x_5 = 0$

What is the new basis?

$$\begin{array}{rcl} x_2 & = & 1 + x_1 - x_3 \\ x_4 & = & 3 - x_1 \\ x_5 & = & 1 - x_1 + x_3 \\ \hline z & = & 1 + 2x_1 - x_3 \end{array}$$

Steps of Simplex algorithm

Step 5 cont'd: Find the entering variable that will improve z

The entering variable $x_1 = 1$ while, $x_2 = 2$, $x_3 = 0$, $x_4 = 2$, $x_5 = 0$

Basis becomes $B = \{1, 2, 4\}$, and $z = 3$

$$\begin{array}{rcl} x_2 & = & 1 + x_1 - x_3 \\ x_4 & = & 3 - x_1 \\ x_5 & = & 1 - x_1 + x_3 \\ \hline z & = & 1 + 2x_1 - x_3 \end{array}$$

Steps of Simplex algorithm

Step 6: Update the simplex tableau

$$x_1 = 1 + x_3 - x_5$$

$$x_2 = 2 - x_5$$

$$x_4 = 2 - x_3 + x_5$$

$$z = 3 + x_3 - 2x_5$$

Can we still improve z??

Steps of Simplex algorithm

Repeat Step 5!: Find the entering variable that will improve z

The objective becomes $z = 3 + x_3 - 2x_5$

Only improving x_3 will improve Z ; thus, the entering variable is x_3

Steps of Simplex algorithm

Repeat Step 5!: Find the entering variable that will improve z

The objective becomes $z = 3 + x_3 - 2x_5$

Only improving x_3 will improve Z ; thus, the entering variable is x_3



How much can you increase x_3 ?

Steps of Simplex algorithm

Repeat Step 5!: Find the entering variable that will improve z

The objective becomes $z = 3 + x_3 - 2x_5$

Only improving x_3 will improve Z ; thus, the entering variable is x_3

? How much can you increase x_3 ?

$$\begin{array}{rclcl}
 x_1 & = & 1 & + & x_3 & - & x_5 & \text{Does not restrict!} \\
 x_2 & = & 2 & & & - & x_5 & \\
 x_4 & = & 2 & - & x_3 & + & x_5 & \\
 \hline
 z & = & 3 & + & x_3 & - & 2x_5 &
 \end{array}$$

$x_3 = 2$

Steps of Simplex algorithm

Step 5 cont'd: Find the entering variable that will improve z

The entering variable $x_3 = 2$ while, $x_1 = 3$, $x_2 = 2$, $x_4 = 0$, $x_5 = 0$

Basis becomes $B = \{1, 2, 3\}$, and $z = 5$

Step 6: Update the simplex tableau

$$\begin{array}{rcl}
 x_1 & = & 3 - x_4 \\
 x_2 & = & 2 - x_5 \\
 x_3 & = & 2 - x_4 + x_5 \\
 \hline
 z & = & 5 - x_4 - x_5
 \end{array}$$

Can we still improve z ??

Steps of Simplex algorithm

Step 5 cont'd: Find the entering variable that will improve z

The entering variable $x_3 = 2$ while, $x_1 = 3$, $x_2 = 2$, $x_4 = 0$, $x_5 = 0$

Basis becomes $B = \{1, 2, 3\}$, and $z = 5$

Step 6: Update the simplex tableau

$$\begin{array}{rcl}
 x_1 & = & 3 - x_4 \\
 x_2 & = & 2 - x_5 \\
 x_3 & = & 2 - x_4 + x_5 \\
 \hline
 z & = & 5 - x_4 - x_5
 \end{array}$$

Can we still improve z ??



No, it's optimal!!